Provenance-Preserving Evidence-Gated Knowledge-Graph Augmentation for Auditable Entity–Relation Extraction

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Abstract

Introduction: Graph-augmented extraction can retrieve useful associations, but a retrieved association is not necessarily a fact stated in the current sentence. Fluent generators also make it difficult to trace a predicted relation to its source span and to audit why it was accepted. This gap is consequential when extracted assertions are used in safety-related knowledge systems.
Methods: We develop a provenance-preserving, evidence-gated framework that separates graph-informed candidate generation from output acceptance. A graph derived only from training annotations supplies bounded concept and relation hints to QLoRA-adapted Qwen3-4B extractors. Deterministic span reconstruction, entity acceptance, and relation verification then retain only source-linked, type-compatible assertions. We evaluate the framework on ADE and CoNLL04 using a shared strict character-span protocol and report the underlying dataset distributions.
Results: The completed test comparisons show that the provenance-gated system improves relation F1 over its matched source-only generator on both benchmarks and records explicit acceptance decisions. The relation improvement is retained in two additional training repetitions on both datasets, although the CoNLL04 entity score does not improve, and after exact CoNLL04 train–test text overlaps are excluded.
Discussion: The results support controlled graph augmentation as an auditable extraction layer for human review. The gates establish structural admissibility and source linkage, making extracted entities and relations more strongly connected to the original text than outputs produced without these constraints, while recording an explicit acceptance path for each assertion, without claiming universal accuracy dominance.

Keywords: entity and relation extraction, knowledge graph, provenance, evidence grounding, adverse drug effects, selective prediction

1. Introduction

Structured information extraction connects unstructured text to searchable entities, typed relations, and downstream knowledge systems. In applications related to safety, an extracted association must also be inspectable: users need to know which mention was identified, which sentence supported the relation, and which transformations produced the final assertion. Drug-related adverse-effect extraction illustrates this requirement directly. A model that recognizes a drug and an adverse effect but invents an association between them can create a misleading record even when its output is syntactically valid. General-domain extraction provides a complementary test of the same technical issue across a broader set of entity and relation labels.

Neural entity and relation models provide effective approaches to the recognition problem [1, 2]. Generative models offer a flexible alternative in which a common structured output format can be reused across label inventories. However, fluent generation does not guarantee that the predicted objects are supported by their input [3]. This distinction matters when extracted triples are inserted into a graph: errors become available to later retrieval or reasoning stages, potentially propagating beyond the sentence that generated them.

Knowledge graphs can contribute typed domain priors [4], and retrieval augmentation can expose those priors without requiring all knowledge to be stored in model parameters [5]. Their use introduces a specific authority problem. A relation observed in a training sentence may suggest what to search for, but it does not establish that the same relation holds in a new sentence. A training graph may also return an example's own annotation or reveal content shared across data splits. The useful question is therefore not only whether graph context increases recall, but whether its eligibility and influence can be controlled and audited.

We address this problem with a staged architecture. Separate source-only, exact-anchor, and high-recall graph-conditioned generators establish a matched comparison of input conditions. A deterministic relation verifier checks structural and source-evidence requirements. A complementary entity gate uses agreement, source-matched anchors, or verified relation endpoints to accept a subset of the graph-conditioned candidates. The final system combines the accepted entities with verified relations whose endpoints survive. The architecture uses existing representation and adaptation methods; its contribution is their governed composition, not a new backbone or a new optimization algorithm.

This paper concentrates on ADE [6] and CoNLL04 [7]. ADE supplies a drug-safety-related extraction task with two entity labels and one relation label. CoNLL04 supplies a general-domain control with four entity labels and five directed relation labels. Both use English sentence-level inputs, making it possible to examine the extraction and verification mechanisms without importing a long-document windowing problem. They are trained and evaluated separately; this is not a cross-dataset zero-shot transfer experiment.

The contributions are threefold:

- A provenance-aware graph-context policy that uses only training annotations, excludes candidates supported only by the current example, and clears graph hints for exact train/evaluation text overlaps.
- A deterministic composition of source-span reconstruction, entity selection, and relation verification that retains evidence-linked assertions and records the reasons behind output decisions.
- A reproducible two-benchmark analysis that combines split and label statistics, uniform strict-span scoring, validation ablations, completed test comparisons, and an exact-overlap sensitivity analysis.

The intended contribution is an auditable extraction layer for analyst-facing knowledge systems. Its primary advantage is to preserve the connection between each extracted assertion and its supporting source text through explicit, inspectable acceptance rules.

2. Related work

2.1. Joint entity and relation extraction

Entity recognition must resolve both mention boundaries and types; relation extraction additionally depends on recovering the correct directed endpoints [1]. Global consistency constraints have a long history in this setting, including the formulation associated with the CoNLL04 benchmark [7]. SpERT enumerates spans and jointly models entities and relations using a pretrained transformer [2]. It is a particularly relevant comparison because it performs the same joint task without requiring a generative JSON interface. Our experiments retrain SpERT from a clean base encoder on the same training partitions rather than using a released task checkpoint that may already include development data.

GLiNER and GLiREL provide label-conditioned alternatives for entity recognition and relation classification [8, 9]. These models differ from locally fine-tuned generators in supervision, pretraining, and output construction. We therefore report the implemented train-calibrated pipeline as a named external baseline, not as an exhaustive assessment of either model family. A same-backbone zero-shot generator further separates the benefit of local adaptation from that of graph conditioning.

2.2. Graph context and constrained generation

Knowledge graphs represent typed entities and relations explicitly, while retrieval augmentation uses selected records as model context [4, 5]. GenIE constrains generative information extraction with a structured output space and entity/relation catalogues [10]. Catalogue validity and source support are different requirements: an admissible entity or relation may still not be asserted in the current input. Our graph context is consequently treated as a candidate prior whose eligibility depends on training provenance and local source matching.

BGE-M3 supplies frozen text representations for semantic pattern retrieval [11]. It is neither a graph neural network nor a jointly optimized component of the extractor. QLoRA enables compact adaptation of the Qwen3 backbone [12, 13]. Neither method alone defines whether a candidate should enter the final assertion graph; the proposed gates make that acceptance policy explicit.

2.3. Evidence and safety-oriented knowledge construction

Hallucination research distinguishes well-formed outputs from source-faithful ones [3]. In safety-related graph construction, prompt-based processing of accident reports likewise illustrates the need to trace generated knowledge back to its sources [14]. ADE offers a more controlled benchmark for a drug-safety-related relation task [6]. The present study uses its released relation annotations to evaluate source-linked extraction and the implemented evidence constraints.

3. Datasets and distribution analysis

3.1. Benchmark sources and partitions

We use the sentence-level ADE and CoNLL04 files distributed in SpERT format. Counts refer to that preprocessed distribution rather than every sentence in the original source corpora. CoNLL04 retains its supplied training, development, and test partitions. The ADE files contain 3,845 training sentences and 427 test sentences; a deterministic development slice of 384 sentences is removed from the supplied training pool, leaving 3,461 training sentences. The slice is selected by a fixed identifier-based procedure, not by validation performance. This is one fixed ADE split, not an average over the multiple folds sometimes used in the literature.

Table 1 gives counts recomputed from the exact imported artifacts. The unit called a document in the software is one benchmark sentence, not a whole medical case report or news article. Both datasets are English; the results therefore support neither multilingual nor cross-lingual claims. Published annotations are imported rather than locally re-annotated.

Table 1: Dataset inventory for the exact experimental partitions. Length is the number of whitespace-delimited tokens in the reconstructed source text.

Dataset	Split	Sentences	Entities	Relations	Mean length
ADE	train	3461	8717	5470	21.17
ADE	validation	384	996	627	21.82
ADE	test	427	1126	724	21.24
CoNLL04	train	922	3377	1283	28.77
CoNLL04	validation	231	893	343	30.27
CoNLL04	test	288	1079	422	28.94

3.2. Entity and relation distributions

ADE contains Drug and Adverse Effect entities and a single Adverse Effect relation label. The imported endpoint convention is *adverse effect to drug*; we preserve it rather than reversing it to a more intuitive drug-to-effect direction. CoNLL04 contains Location, Organization, Person, and Other entities, with Work For, Kill, Organization Based In, Live In, and Located In relations.

The training-label distributions are summarized in the accompanying tables, and Table 2 lists exact label counts for every partition. The single

ADE relation label eliminates inter-class relation confusion but does not eliminate endpoint detection, direction, or false-pair errors. CoNLL04 introduces a broader relation inventory and more heterogeneous entity roles. Consequently, a higher aggregate F1 on ADE should not be interpreted as evidence that drug-safety reasoning is inherently easier than general-domain reasoning; the supervised extraction tasks differ in structure and size.

Table 2: Exact entity and relation annotation counts in the imported files.

Dataset	Kind	Label	Train	Validation	Test
ADE	Entity	ADVERSE EFFECT	4631	529	616
ADE	Entity	DRUG	4086	467	510
ADE	Relation	ADVERSE EFFECT	5470	627	724
CoNLL04	Entity	LOCATION	1219	322	427
CoNLL04	Entity	ORGANIZATION	616	170	198
CoNLL04	Entity	OTHER	455	118	133
CoNLL04	Entity	PERSON	1087	283	321
CoNLL04	Relation	KILL	179	42	47
CoNLL04	Relation	LIVE IN	330	91	100
CoNLL04	Relation	LOCATED IN	247	65	94
CoNLL04	Relation	ORGANIZATION BASED IN	271	76	105
CoNLL04	Relation	WORK FOR	256	69	76

3.3. Sentence lengths and coverage

The mean training lengths are 21.17 reconstructed tokens for ADE and 28.77 for CoNLL04. Their test means are 21.24 and 28.94, respectively. The cumulative distributions describe all three partitions rather than only the training set. These lengths are descriptive whitespace-token counts, not Qwen tokenizer counts. The largest observed reconstructed length is 97 for ADE and 118 for CoNLL04 across the imported partitions.

Every imported sentence contains at least one annotated relation. This is an important property of the selected benchmark files: they do not measure false-positive behavior over a deployment stream dominated by sentences with no relevant relations. We retain the supplied benchmark composition and do not create additional negative sentences for the reported tests.

3.4. Split overlap and graph-context isolation

The graph is built separately for each dataset from its training annotations only. An exact-overlap check compares case-folded source strings after collapsing whitespace. ADE has no train/development or train/test exact-text overlap under this check. CoNLL04 has 14 development sentences and 19 test sentences whose text matches training content. For these sentences, both exact-anchor and high-recall graph context are cleared before generation.

Clearing graph hints does not undo exposure of the trained generator to duplicate training text. We therefore retain the supplied partitions for the main comparison and report an additional fixed-prediction test analysis that excludes the 19 overlapping CoNLL04 test sentences. We do not claim that the benchmark is duplicate-cluster isolated or that this exact-match check detects all near duplicates.

4. Proposed method

4.1. Source-grounded representation

Let a sentence x be the deterministic single-space concatenation of its released tokens. An entity mention is represented as

$$e = (b, u, y), \qquad x[b : u] = m, \tag{1}$$

where $[b, u)$ is a character interval, y is an entity label, and m is the copied mention text. A relation $r = (e_s, z, e_t)$ has typed directed endpoints and relation label z. Source identifiers and evidence spans are stored with these objects. Repeated strings at different offsets remain distinct mentions. The character-span requirement in Equation (1) prevents a normalized name alone from receiving credit for the wrong occurrence.

The imported benchmark assigns every relation the status `explicit` and uses its source sentence as the evidence span. This representation keeps each evaluated relation tied to a concrete source sentence and makes the evidence path available for inspection.

4.2. Architecture and comparison systems

Figure 1 summarizes the implemented flow. SOE (Source-Only Extraction), EAE (Exact-Anchor Extraction), and HRGE (High-Recall Graph Extraction) are separately trained QLoRA adapters on the same Qwen3-4B snapshot. SOE uses source text and the training-derived ontology without graph hints. EAE receives conservative concept hints, while HRGE receives bounded anchors, edge priors, and semantic relation patterns.

EVGE (Evidence-Verified Graph Extraction) applies the relation verifier to HRGE. CFE (Conservative-Fusion Extraction) pairs accepted HRGE entities with endpoint-filtered raw HRGE relations. PGE (Provenance-Gated Extraction) pairs the same accepted entities with endpoint-filtered verified relations. These three systems are deterministic post-processing variants, not additional trained networks. In deployment PGE needs EAE and HRGE candidates; SOE is a comparison branch and does not participate in fusion.

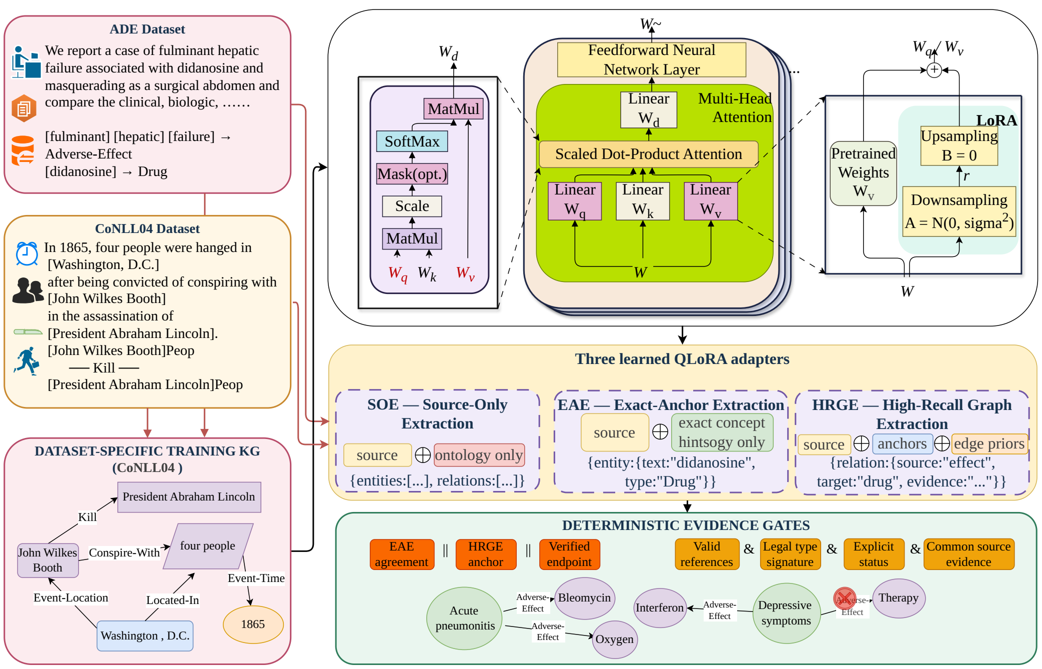

Figure 1: Detailed methodology used in this study. The figure shows the ADE and CoNLL04 inputs, dataset-specific training knowledge graphs, the Transformer attention block, the SOE/EAE/HRGE QLoRA adapter branches, and deterministic evidence gates based on agreement, anchors, endpoints, relation validity, and source evidence.

4.3. Provenance-controlled graph context

Each training graph stores mention strings, types, directed relations, and the sentence identifiers supporting them. For a training sentence d, a candidate is excluded if its only supporting provenance is d. Candidates with other supporting sentences remain eligible; their stored aggregate support statistics are not recomputed as a fully leave-one-out graph. This distinction prevents interpreting the provenance policy as complete removal of every contribution from the current example.

EAE uses type-balanced, exact-matching concept hints. HRGE uses three context channels. First, source-matched anchors must pass type-purity and mention-form filters. Second, a typed edge prior requires support outside the current example and both endpoint strings in the current source. Third, frozen BGE-M3 retrieves semantically related relation-pattern records. If $\mathbf{H}$ contains normalized eligible record embeddings and $\mathbf{q}$ is the normalized source embedding, retrieval scores are

$$\mathbf{s} = \mathbf{H}\mathbf{q}, \qquad \mathcal{I} = \mathrm{TopK}_k\{i : s_i \geq \tau\}. \tag{2}$$

Equation (2) ranks candidate context, not accepted facts. The bounded retrieval limits and thresholds are fixed in the protocol artifact rather than tuned on the test set. The prompt explicitly identifies retrieved objects as hints. Exact-overlap quarantine overrides retrieval and clears all graph context for the affected evaluation sentences.

4.4. Compact generation and deterministic reconstruction

The generator emits entity identifiers, copied text and types, plus relation identifiers, directed endpoint references, types, and explicit status. It is not asked to calculate source offsets. Deterministic expansion assigns copied mentions to occurrences in the supplied sentence and reconstructs character spans; unmatched strings and invalid references are recorded. This separates language-model candidate generation from source-coordinate bookkeeping.

All three generators use one-epoch QLoRA adaptation of the same frozen Qwen3-4B backbone. Low-rank updates are applied to the attention and feed-forward projections, while prompt tokens are excluded from the training loss. The same dataset partition and optimization configuration are used for all branches and repetitions; full settings are retained in the artifact manifest rather than repeated in the narrative.

Inference uses the full supplied sentence and a fixed generation policy. Failed jobs are represented by empty predictions, not removed from the denominator. These experiments do not use annotation-guided rescue windows or label-dependent sentence truncation.

4.5. Relation verification and entity acceptance

For each relation label z, training annotations define permitted source-type and target-type sets S_z and T_z. The implemented type constraint is $y_s \in S_z$ and $y_t \in T_z$; it does not additionally require that every specific source/target type pair has been observed together. The verifier $V(r)$ also requires existing distinct entity references, a known relation label, explicit status, valid evidence, and both endpoint strings in a common source-evidence span. Its accepted set is

$$R_V = \{r \in R_H : V(r) = 1\}. \tag{3}$$

The entity gate uses three Boolean signals:

$$A(e) = A_{\text{agree}}(e) \vee A_{\text{anchor}}(e) \vee A_{\text{endpoint}}(e), \quad E_A = \{e \in E_H : A(e) = 1\}. \tag{4}$$

Agreement means that EAE predicts the same case-folded, whitespace-normalized text and type. Anchor support requires a same-type source-gated HRGE anchor. Endpoint support means that the entity participates in a relation retained by Equation (3). The three signals in Equation (4) are complementary but not statistically independent; verified endpoints are a rule-based signal, not a second human judgement. Duplicate normalized text/type candidates are collapsed within the sentence.

Let $F(R, E)$ retain relations in R only when both referenced entities survive in E. The two compositions are

$$G_{\text{CFE}} = (E_A, F(R_H, E_A)), \qquad G_{\text{PGE}} = (E_A, F(R_V, E_A)). \tag{5}$$

Equation (5) makes the difference between raw-relation fusion and verified fusion explicit. Accept/reject decisions are logged. A retained relation satisfies the stated structural and source-linkage rules, and the applied checks are recorded with the output for inspection.

5. Experimental protocol

5.1. Research questions and scope

We ask whether PGE improves strict entity and relation extraction over SOE; how graph context and deterministic selection contribute on development data; and whether observed gains persist after exact-overlap sentences are removed. External baselines provide a direct reference for the generative design and help separate source-linkage gains from changes in the underlying extractor.

The two-benchmark scope is a focused retrospective analysis rather than a preregistered representative sample of extraction domains. Model and gate settings were fixed before the reported test evaluation. Two additional complete repetitions provide a limited stability check for SOE and PGE on the same datasets.

5.2. Baselines and evaluation parity

The internal development comparison includes all six named systems. The test table includes the two promoted systems and three completed external comparisons. Qwen3-4B zero-shot uses the same backbone without local adapters, followed by the recorded source expansion and relation verifier. SpERT is trained from a base encoder on the same training partitions and its final checkpoint is used without selecting a test-best epoch. GLiNER+GLiREL uses predicted, not gold, entities and training-only calibration. Detailed baseline hyperparameters are retained in the experiment artifacts. These are heterogeneous baseline configurations, not equal-compute experiments.

All reported main-table predictions are rescored with one public character-span evaluator. An entity key contains its exact source interval and type. A relation key contains both typed directed entity keys and the relation label. Predicted evidence offsets must resolve to the supplied source; the evaluator does not recover missing offsets by searching normalized text. Duplicate keys receive one vote, invalid objects remain unmatched predictions, and omitted jobs remain empty. For pooled counts,

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}, \quad F_1 = \frac{2TP}{2TP + FP + FN}. \tag{6}$$

Equation (6) is applied across sentences within each dataset, not after averaging their individual F1 scores. Because all imported statuses are explicit, claim-status-aware scores do not form a separate uncertainty benchmark. The unified evaluator removes small denominator differences caused by legacy duplicate handling; unchanged source artifacts and a reconciliation report preserve the earlier metrics.

5.3. Statistical and compliance analyses

We compute paired bootstrap resamples of test sentences from fixed predictions and recompute pooled F1 within each resample. The percentile interval describes sentence-sampling uncertainty for the completed model pair, not variability across training repetitions. Intervals are descriptive, unadjusted for multiple comparisons, and are not used to change the frozen models or gates.

For the two additional training repetitions, we report the arithmetic mean and sample standard deviation of strict-span test F1. With only two repetitions, these summaries show observed initialization sensitivity but do not reliably estimate population variance. Per-step training loss was not retained by the formal matrix, so no reconstructed or interpolated loss curve is reported.

Evidence coverage and rule-invalid counts are measured separately from extraction F1. These compliance counts refer to emitted prediction objects before the evaluator's key deduplication. Exact-overlap sensitivity removes the same affected test sentence identifiers from both systems, without retraining or regenerating predictions. Test labels are used for reporting statistics and scoring only, not for constructing graph hints or choosing thresholds in this revision.

6. Results

6.1. Completed test comparisons

Table 3 reports the completed test results under the common evaluator. PGE raises ADE relation precision from 80.62% to 87.00% and recall from 67.82% to 72.10%, giving relation F1 of 78.85% compared with 73.67% for SOE. Entity F1 remains essentially unchanged, at 87.93% versus 87.97%. Thus, the main improvement is relational rather than a general gain in mention recognition.

On CoNLL04, PGE raises relation precision from 54.61% to 62.40% and recall from 56.16% to 57.82%. Relation F1 increases from 55.37% to 60.02%, while entity F1 decreases from 83.83% to 83.20%. The composition trades some entity coverage for a better relation precision–recall balance in this run.

SpERT has the highest entity and relation F1 among these completed systems: 91.37%/84.08% on ADE and 89.02%/69.72% on CoNLL04. PGE therefore improves the matched generative baseline but does not surpass this trained span-based model. PGE also exceeds the recorded zero-shot and GLiNER+GLiREL pipelines on both extraction tasks. The latter observation is specific to their implemented label mapping and calibration; it is not a claim that every configuration of those model families would perform worse.

Table 3: Completed test comparisons, strict source-character-span micro scores (%). Qwen3-ZS is Qwen3-4B zero-shot with the recorded verifier; GLiNER+GLiREL uses training-only calibration. Each dataset uses its own test partition. All rows are rescored with the same evaluator.

Dataset	System	Ent. P	Ent. R	Ent. F1	Rel. P	Rel. R	Rel. F1
ADE	Qwen3-ZS + verifier	63.83	63.32	63.58	39.36	22.24	28.42
ADE	GLiNER + GLiREL	79.30	64.65	71.23	60.31	43.23	50.36
ADE	SpERT	90.65	92.10	91.37	82.19	86.05	84.08
ADE	SOE	92.21	84.10	87.97	80.62	67.82	73.67
ADE	PGE	92.89	83.48	87.93	87.00	72.10	78.85
CoNLL04	Qwen3-ZS + verifier	48.78	53.94	51.23	30.92	19.19	23.68
CoNLL04	GLiNER + GLiREL	60.78	14.37	23.24	13.79	3.79	5.95
CoNLL04	SpERT	88.61	89.43	89.02	71.01	68.48	69.72
CoNLL04	SOE	85.83	81.93	83.83	54.61	56.16	55.37
CoNLL04	PGE	86.04	80.54	83.20	62.40	57.82	60.02

6.2. Additional-run stability check

Table 4 summarizes the two later repetitions. On ADE, PGE improves relation F1 over SOE in both repetitions (81.20% versus 78.31%, and 80.62% versus 78.06%). On CoNLL04, the corresponding differences are positive but smaller (58.50% versus 56.91%, and 61.45% versus 60.92%). Across the two runs, the mean relation-F1 difference is 2.73 points on ADE and 1.06 points on CoNLL04. CoNLL04 entity F1 decreases by 0.51 points on average, so this check supports a relation-specific rather than universal improvement.

Table 4: Test stability across two additional training repetitions under the common strict-span evaluator (micro %, mean ± sample standard deviation; $n = 2$). The final column is the mean PGE-minus-SOE relation-F1 difference.

Dataset	System	Entity F1	Relation F1	Δ relation F1
ADE	SOE	88.00 ± 0.37	78.18 ± 0.18	–
ADE	PGE	89.35 ± 0.04	80.91 ± 0.41	+2.73
CoNLL04	SOE	82.26 ± 1.91	58.92 ± 2.84	–
CoNLL04	PGE	81.75 ± 2.02	59.98 ± 2.09	+1.06

6.3. Paired differences and uncertainty

Using unrounded counts, PGE-minus-SOE relation F1 is 5.18 percentage points on ADE, with a paired 95% interval of [0.31, 10.29], and 4.65 points on CoNLL04, with an interval of [0.04, 9.12]. The positive point estimates are consistent with the test-table improvements, but the intervals are broad and their lower bounds are close to zero. We interpret them as evidence about these completed test predictions, not as a general guarantee across training runs. Entity-F1 differences are −0.04 points on ADE, interval [−1.75, 1.62], and −0.63 on CoNLL04, interval [−2.38, 1.04]; neither supports an entity-F1 improvement claim.

6.4. Development-set component analysis

Table 5 separates candidate generation from deterministic selection on development data. On ADE, HRGE has relation F1 of 76.29%, above SOE's 73.16%. EVGE is identical to HRGE, and PGE is identical to CFE, showing that the relation verifier removes no additional relations in these variants. The entity gate reduces HRGE recall and gives PGE relation F1 of 75.91%. It is therefore incorrect to attribute all ADE gains to verification or to claim that every stage independently improves F1.

On CoNLL04, HRGE relation F1 is 58.26%, below SOE's 60.24%. Verification raises it to 59.60%, and the final composition reaches 60.19%. This is effectively unchanged from SOE at the development point estimate, rather than a clear development-set gain. Entity precision rises after gating, while recall and F1 decrease. The distinction between development and test behavior is retained rather than replacing the ablation with a test-selected variant.

Table 5: Development-set ablation under the common strict-span evaluator (micro %). These rows characterize the fixed architecture and are not additional promoted test systems.

Dataset	System	Ent. P	Ent. R	Ent. F1	Rel. P	Rel. R	Rel. F1
ADE	SOE	89.25	82.53	85.76	78.82	68.26	73.16
ADE	EAE	90.49	81.22	85.61	80.84	67.30	73.46
ADE	HRGE	90.35	85.54	87.88	80.04	72.89	76.29
ADE	EVGE	90.35	85.54	87.88	80.04	72.89	76.29
ADE	CFE	90.48	83.94	87.08	81.39	71.13	75.91
ADE	PGE	90.48	83.94	87.08	81.39	71.13	75.91
CoNLL04	SOE	84.94	82.08	83.49	61.33	59.18	60.24
CoNLL04	EAE	83.71	82.87	83.29	58.10	60.64	59.34
CoNLL04	HRGE	82.35	81.52	81.94	60.06	56.56	58.26
CoNLL04	EVGE	82.35	81.52	81.94	62.99	56.56	59.60
CoNLL04	CFE	84.97	78.50	81.61	63.37	55.98	59.44
CoNLL04	PGE	84.97	78.50	81.61	65.08	55.98	60.19

6.5. Rule compliance and its interpretation

The test SOE output contains 31 rule-invalid relation objects among 610 ADE objects and 16 among 436 CoNLL04 objects. PGE retains 600 and 392 relation objects, respectively, with zero rule-invalid objects. HRGE already has zero invalid ADE relations and only three invalid CoNLL04 relations among 413 objects. These counts distinguish the contribution of candidate quality from that of the verifier.

The measured unsupported-evidence count is zero for SOE, HRGE, and PGE on both test datasets. Because each relation's evidence is the whole sentence, co-occurrence is easy to satisfy once endpoints resolve. Thus, these experiments do not demonstrate a reduction from a nonzero unsupported-claim rate. They show explicit evidence binding and rule compliance, while the gold-referenced F1 scores measure a different property: whether the predicted typed relation agrees with the annotation.

6.6. Exact-overlap sensitivity

Removing the 19 exact train/test overlaps leaves 269 CoNLL04 test sentences, 1,023 gold entities, and 403 gold relations. SOE relation F1 becomes 53.79% and PGE becomes 59.10%, a 5.31-point difference; entity F1 is 83.02% and 82.90%, respectively. The observed relation advantage therefore persists without those exact overlaps. This analysis addresses exact repeated text, not all forms of semantic overlap or pretraining contamination. ADE has no such overlaps, so its sensitivity result is identical to the full test.

7. Discussion

7.1. What the two benchmarks establish

The common result is a relation-F1 improvement over source-only generation. The gains arise alongside higher relation precision on both tests, while entity F1 stays similar or declines slightly. This is consistent with an architecture designed to govern candidate acceptance rather than simply maximize output volume. The validation ablation further shows that graph conditioning and selection interact: recall-oriented context can help on ADE but does not automatically improve CoNLL04, and a verifier may be inactive when candidates already satisfy its rules.

The distribution analysis helps interpret these differences. ADE has a larger training pool, shorter average sentences, and a single relation class. CoNLL04 has fewer training sentences, longer inputs, and five relation labels. These factors are confounded rather than experimentally isolated. Their association with the observed scores motivates separate dataset reporting; the factors should not be conflated in a single pooled interpretation.

7.2. *Auditability as a distinct design objective*

A system exposes a useful acceptance mechanism for generated graph objects. The contribution is strongest when the entire path is inspectable: graph support identifies why a candidate was suggested; reconstructed offsets identify where it appears; gate records identify why it was retained or rejected. However, this benefit is architectural. No user study in the present experiments measures reduced analyst effort, improved decisions, or greater trust. It would be inappropriate to substitute perfect rule compliance for such evidence.

The gates also impose a coverage cost. Agreement operates on normalized text/type pairs, and repeated mentions may be collapsed during fusion even though the evaluator distinguishes offsets. This can remove legitimate occurrences. A legal endpoint signature and sentence-level co-occurrence are insufficient to prove a relation's meaning, and three gate signals should not be interpreted as three independent confirmations. These observations explain why the final output remains a candidate assertion graph for review.

7.3. *Implications for safety-related applications*

ADE links the experiments to drug-safety information extraction, while CoNLL04 tests general relational structure. The framework can present an analyst with the candidate relation, source sentence, typed endpoints, and provenance of its acceptance path. An analyst can then accept, amend, or reject the proposed association. Absence from the retained graph is not evidence of the absence of an adverse effect or another real-world relation.

The reported task focuses on source-linked extraction and evidence-path inspection. The framework is intended to support analyst review by making the origin and acceptance path of each extracted assertion explicit.

8. Limitations and responsible use

The focused, retrospectively narrowed dataset scope and small number of completed training repetitions limit generalization. The fixed ADE partition is not directly interchangeable with published multi-fold results. Both corpora are English and consist of relation-bearing sentences; multilingual behavior, long-report processing, and false-positive rates on unrestricted text streams are outside the evaluated setting. Sentence-level bootstrap intervals cannot recover dependence between sentences originating from the same upstream article when those article identifiers are unavailable.

Exact-overlap quarantine prevents direct graph hints for known repeated source text but does not remove the generator's training exposure. The sensitivity analysis excludes known exact duplicates only. Frozen pretrained encoders and generators may also have seen related material during pretraining, which the available artifacts cannot rule out. The ontology's allowed source and target sets are less restrictive than a catalogue of observed joint type pairs.

The one-epoch generative systems, 20-epoch SpERT, and pretrained label-conditioned models use different training schedules. PGE requires two adapted generators plus frozen retrieval, so compact adaptation must not be confused with measured end-to-end cost savings. No new clinical annotations or independent evidence-judgement labels were collected for this paper. Public availability of a dataset or model does not by itself permit every downstream redistribution or commercial use; original licenses remain applicable.

9. Conclusions

This study presents a provenance-preserving, evidence-gated graph-augmentation framework and evaluates it on ADE and CoNLL04. Dataset statistics expose the label and input-length differences underlying the two benchmarks. Under a unified strict-span evaluator, the completed PGE tests improve relation F1 over SOE by 5.18 and 4.65 percentage points, respectively, while offering an explicit source-linked acceptance path. The CoNLL04 relation advantage remains after exact train/test overlaps are excluded, and two additional runs retain positive, though smaller on CoNLL04, relation-F1 differences. The supported outcome is a controlled, auditable graph-informed extraction layer that keeps generated entities and relations strongly connected to their source text and exposes the acceptance path for human review.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence this work.

Data and code availability

The experiments use the public ADE and CoNLL04 distributions in SpERT format. An anonymized reproducibility package will contain the split construction, dataset statistics, source hashes, evaluation code, and derived result tables. Dataset text and model weights remain subject to their original licenses. The identifying repository URL is withheld during double-blind review.

Declaration of generative AI and AI-assisted technologies in the writing process

The authors used OpenAI Codex to assist with code inspection, language editing, figure preparation, and manuscript organization. The authors are responsible for reviewing the resulting material and for the final content. The language models evaluated as research systems are described separately in the Methods.

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